

Population genomics of the northern shortfin squid *Illex illecebrosus*: strong expansion, a balancing-selected chromosome-2 inversion, and a genome-wide scan for selective sweeps

[Author One]¹ and S. Small^{1,*}

¹Institute of Ecology and Evolution, University of Oregon, Eugene, OR, USA

*Correspondence: ssmall@uoregon.edu

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Abstract

The northern shortfin squid *Illex illecebrosus* is a short-lived, highly fecund, broadcast-spawning cephalopod that supports major commercial fisheries across the North Atlantic, yet its genome-scale population history and the targets of natural selection are unknown. We analysed a 350-individual, whole-genome callset spanning 45 autosomes and the Z chromosome. The population carries the signature of a strong, ongoing demographic expansion — genome-wide nucleotide diversity $\pi \approx 0.0093$ with Tajima's $D \approx -2$ in essentially every window — and a moments-fit history of a ~ 12 -fold size increase from an ancestral $N_e \approx 5.5 \times 10^5$ to a present $N_e \approx 6.8 \times 10^6$. Consistent with a recent population-genomic survey (Baker et al., 2025), population structure is confined to a single large inversion on chromosome 2; here we go beyond that description to establish what the inversion *is* and to map the genome-wide landscape of recent positive selection. The chr2 inversion (chr2:60.54–79.50 Mb, ~ 19 Mb) drives all apparent structure ($F_{ST} = 0.37$ between arrangements, geographic $F_{ST} \approx 0$), is non-clinal, and is *not* neutral: two mutation-rate-free arguments and a coalescent fit with an ARG simulator that enforces the arrangement barrier show it arose ~ 850 ka, was maintained near a frequency of ~ 0.74 for most of its history, and has been actively removed to its present 0.37 over the last ~ 50 – 100 ky under selection of order $N_e s \sim 10^2$ – 10^3 . We further build a per-sex recombination map, quantify moderate gene-density-driven background selection, identify candidate adaptive genes by McDonald–Kreitman tests, and perform a genome-wide selective-sweep scan. Because a simulation-trained per-window classifier over-calls sweeps on real, expansion-shaped, missing-data-rich genomes, we deploy it as a stratified empirical-outlier detector corroborated by a second method, yielding a small, biologically coherent candidate set — including neural and RNA-regulatory genes, a cephalopod-lineage-specific gene under a hard sweep, and, on the Z chromosome, a cluster of RNA-regulation and hormone-signalling loci. Our results establish *Illex illecebrosus* as a large- N_e , expansion-dominated system in which a single balancing-selected supergene is the sole axis of structure, and provide a reusable, honestly-calibrated framework for sweep detection under these conditions.

1 Introduction

Cephalopods combine an unusual life history — semelparity, enormous fecundity, near-total larval mortality, and boom–bust recruitment — with population sizes among the largest of any animal.

The northern shortfin squid, *Illex illecebrosus*, is an annual, semelparous broadcast spawner distributed across the northwest Atlantic shelf and slope, where it supports large and economically important fisheries. These traits make it a compelling but under-studied system for population genomics: extreme fecundity with high early mortality is the demographic profile that motivates sweepstakes reproductive-success models, in which a few individuals contribute disproportionately to each generation, while a large effective size sharpens the resolution of both demographic inference and selection scans.

A recent population-genomic survey of *Illex illecebrosus* (Baker et al., 2025) established the population-structure picture we confirm here — weak overall differentiation with a single, striking axis of structure driven by a chromosome-2 inversion, recovered by PCA, F_{ST} , and admixture analysis. That description leaves open the questions that motivate this study: *why* the inversion segregates as it does, and *where else* in the genome selection has acted. We therefore focus on two things that a structure-and-differentiation survey does not resolve. First, the evolutionary character of the chr2 inversion: is it a neutral, drifting polymorphism or a selected supergene, and if selected, of what kind, what age, and with what frequency history? Second, the genome-wide landscape of recent positive selection — the selective sweeps, on the autosomes and the sex chromosome, that a differentiation scan cannot see. Underpinning both, we ask whether *Illex illecebrosus*’s demography can be distinguished from the multiple-merger coalescent that its sweepstakes reproduction would predict, since the answer licenses every age and selection estimate that follows.

We address these with a whole-genome dataset of 350 *Illex illecebrosus* individuals, supported by an *I. coindetii* outgroup for divergence-based inference. We infer demography from the site-frequency spectrum and linkage disequilibrium, dissect the chr2 inversion with per-arrangement diversity, divergence, and a purpose-built coalescent model of a barrier-bearing chromosome, build a per-sex recombination map, quantify background selection, and scan for selective sweeps. Throughout, we treat the mismatch between simulated and real neutral genomes not as a nuisance to be hidden but as a finding in its own right that dictates how a sweep scan must be framed.

2 Results

2.1 Sex determination and genome organisation

We genotyped 633 *Illex illecebrosus* individuals sampled across eleven NAFO management divisions (fig. 1) spanning ~ 37 – 50°N (table 1). Coverage on the Z chromosome cleanly separates the sample into **342 ZZ males and 291 ZO females**, consistent with a ZZ/ZO sex-determination system in which females are the heterogametic sex, carrying a single Z with no W chromosome. A chromosome previously suspected to be sex-linked (chr42) is autosomal by this criterion and is treated as such throughout. All autosomal analyses use both sexes; the Z chromosome, on which males are diploid and females haploid, is analysed separately (section 2.8).

2.2 Population structure is confined to the chromosome-2 inversion (confirming prior work)

Consistent with Baker et al. (2025), a windowed principal-component analysis finds population structure in exactly one place in the genome: inside the chromosome-2 inversion, where the among-cluster variance explained is $\eta^2 = 0.748$, against $\eta^2 = 0.003$ in collinear sequence and on every other chromosome (fig. 2). There is no cryptic geographic or cryptic-lineage structure elsewhere, and the residual dispersion along the minor PCA axes is attributable to a handful of individuals

rather than to any population axis (it tracks neither division nor latitude nor coverage). A KING-robust relatedness screen of the autosomal genotypes further finds **no cryptic family structure** — no duplicate, parent–offspring, or full-sib pairs and a single second-degree pair among the 633 individuals (maximum pairwise kinship 0.16) — so despite the sweepstakes reproductive mode the sample behaves as an effectively unrelated set of individuals, and relatedness cannot inflate the diversity or demographic inferences below. Because a per-SNP differentiation scan can detect local adaptation that the diversity-based sweep tests would miss — a sensitive test in a population this panmictic — we also compared per-SNP F_{ST} among divisions to a coalescent neutral null (a single panmictic population simulated under the fitted demography at the empirical sample sizes and missingness): across the full callsets (551,344 SNPs covering 12,007 genes) the empirical differentiation lies entirely within the drift expectation — *no* SNP exceeds the neutral maximum, a single intergenic outlier passes the lenient threshold against ≈ 556 expected by chance ($\sim 555 \times$ fewer than the false-positive rate), and there are *zero* genic outliers — so there is no signal of geographic local adaptation anywhere in the genome, and the chr2 inversion is geographically non-clinal (Supplementary Figs. S2, S19). We treat this as an established starting point rather than a result: it confirms the earlier PCA/ F_{ST} /admixture picture at per-window resolution, means population structure cannot be invoked as a confounder for the analyses that follow, and — most importantly — reframes the inversion from “the thing that structures the population” into an evolutionary object to be characterised in its own right (section 2.3).

2.3 A large, balancing-selected inversion on chromosome 2

The chr2 inversion is the central feature of the *Illex illecebrosus* genome. Local-PCA and a Gaussian-mixture karyotype assignment (95.3% confident) resolve three genotype clusters — 254 AA, 284 AB, and 95 BB individuals — whose differentiation is karyotypic, not geographic: $F_{ST}(AA, BB) = 0.37$ within the inversion against a geographic $F_{ST} \approx 0$. The three karyotypes co-occur throughout the sampled range, with no latitudinal cline in arrangement frequency (0.29–0.46 across divisions; Supplementary Fig. S4 and Table S2). Two independent lines pin the breakpoints to **chr2:60.54–79.50 Mb** (a ~ 18.96 Mb differentiated body): a step-change in arrangement differentiation and a strand flip in a whole-chromosome alignment to *I. coindettii*, which independently confirm the right breakpoint. The nominal 60.0–80.0 Mb extent overshoots by ~ 0.5 Mb at each end.

Polarisation. The derived, inverted arrangement is **BB, at a present frequency of 0.374**; AA is ancestral and standard. This polarisation is the reverse of a naive one based on the reference allele, which is circular because the *Illex illecebrosus* reference assembly is itself in the inverted orientation (it carries BB at 86.5% of fixed-different sites). Restricting to the non-circular subset of sites where *I. coindettii* carries the *Illex illecebrosus* alternate allele shows AA to be ancestral (63.0%) and BB derived — a single inversion event, with the reference simply assembled from an inverted individual.

The inversion is not neutral. The primary evidence is per-arrangement diversity (table 2, fig. 3). The rare derived arrangement is *more* diverse than its current frequency can sustain: $\pi_I/\pi_S = 1.356 \pm 0.048$ against an equilibrium expectation of $p/(1-p) = 0.60$ for a polymorphism held at 0.374 — a 2.3-fold *excess*, the signature of an old polymorphism that was once commoner, not a recent rise. Two mutation-rate-free diffusion arguments exclude neutrality. First, *attainment*: reaching the inferred historical peak frequency of ~ 0.74 by drift takes at minimum ~ 1.15 million generations at the binding (smallest) N_e , i.e. $\sim 1.36 \times$ the entire inferred lifetime of the inversion, and up to $17 \times$ at present N_e . Second, the *decline*: the observed fall of $0.74 \rightarrow 0.374$ is 6.7 standard

deviations of neutral drift at the fitted duration, and a coalescent profile of the fall duration excludes the slow, drift-compatible alternative at $\Delta\chi^2 \gtrsim 8\text{--}80$. The arrangement was actively removed, not lost to drift.

Age and history. We fit the inversion’s age and frequency trajectory with **msinv**, an ARG-based coalescent simulator that enforces the cross-karyotype recombination barrier and outputs tree sequences (section 5). Conditioning on a deterministic frequency trajectory under within-arrangement neutrality — the model the data support, since they constrain the *shape* of $p(t)$, not a site-level mechanism — the joint fit to π_I/π_S , d_{xy}/π_I , and the per-arrangement singleton spectrum (fig. 4; Supplementary Fig. S3) gives a coalescent age of **~ 850 ky**. The age is the robustly-determined quantity because it is carried by d_{xy} between arrangements, which is symmetric to the polarisation. The reconstructed history is: the inversion arose ~ 850 ka, coincident with the onset of the population expansion; was maintained near $p \approx 0.74$ for most of its history; and then fell sharply to 0.374 over $\sim 50\text{--}100$ ky ending $\gtrsim 50\text{--}100$ ka, under selection of order $N_e s \sim 10^2\text{--}10^3$ ($s \approx 1.5 \times 10^{-4}$). Every absolute age scales inversely with the mutation rate $\mu = 3 \times 10^{-9}$, which remains the dominant external uncertainty.

What the inversion is not. Recombination inside the inversion is at the chromosomal background rate by direct measurement (the ~ 19 kb LD windows a recombination-map method reads are blind to the 19-Mb barrier, so the interior rate is not a barrier readout), so the differentiation is a consequence of the barrier, not of locally suppressed crossing-over. The region is *gene-poor* (30 annotated genes in the differentiated body against a median of 50 in random windows of the same size), its annotation is demonstrably incomplete ($\sim 12\text{--}14$ genes missed, recovered by cross-species protein homology), and a previously-reported metabolic Gene-Ontology enrichment does not survive a linkage-aware null. No single locus stands out in coding sequence, regulatory compartments, per-transcript divergence, or a karyotype-stratified sweep scan: the block is differentiated as a unit, which is what a recombination barrier produces. Gene flux between arrangements is not detectable in mean d_{xy} , though a residual $\sim 40\%$ depletion of near-fixed differences in the outermost megabase remains after the breakpoints are pinned. The mechanism of balancing selection — overdominance, associative overdominance, frequency dependence, or a moving optimum under fluctuating selection — is not resolved by these data; a shelf-spawning squid whose paralarvae ride the Gulf Stream is a plausible setting for temporally fluctuating selection on ~ 100 ky glacial cycles, but this remains a hypothesis.

Gene content and the architecture of arrangement divergence. The block has no functionally coherent cargo and no direct signal of positive selection, yet it carries the molecular signature of a young, recombination-suppressed supergene. The body carries 89 gene models, 45 with cross-species functional annotation (Supplementary Table S7); it is gene-poor and functionally scattered (lipid metabolism, DNA repair, transcription/splicing, signalling), with no coherent pathway. A gene-ontology test of its annotated genes against the genome returns *no* enriched term at $\text{FDR} < 0.05$ in any ontology, and a permutation null confirms this is genuine: the apparent nominal signal (and the eye-catching ~ 40 -fold per-term enrichments, which rest on as few as two genes) is smaller than what random gene sets of the same size produce (69 vs 86.6 ± 60 nominal terms; permutation $p = 0.51$; Supplementary Fig. S21). The inversion has thus captured a gene-poor, functionally unremarkable slice of the genome, not a selected gene set. What the arrangements *do* accumulate is divergence-as-a-unit: between-karyotype F_{ST} is ≈ 0 across collinear chromosome 2 but jumps to a median of 0.34 inside the inverted body (Supplementary Fig. S20), where 3,513

near-fixed differences between the AA and BB arrangements (per-arrangement $|\Delta AF| \geq 0.9$) accumulate (Supplementary Fig. S5). These are overwhelmingly non-coding — coding sequence is only $\sim 0.2\%$ of the region — but the exonic fixed differences are ~ 2.3 -fold over-represented relative to that footprint and are strikingly *clustered*: all 19 exonic fixed differences fall in just nine genes, led by a cytochrome P450 (*CYP4X1*) and an acidic-repeat protein (four each) and including a phospholipase A2, the ciliary gene *RPGRIP1L*, and a cysteine-rich venom-like protein. Thus, although no locus in the block shows a classical sweep and the region is enriched for no function, the coding divergence that recombination suppression shelters is concentrated in a handful of genes rather than spread evenly — the expected architecture of a balancing-selected inversion that recombines with its homologue only rarely.

2.4 Demographic history: strong expansion, not multiple mergers

Genome-wide, *Illex illecebrosus* shows $\pi \approx 0.0093$ and Tajima’s $D \approx -2$ in essentially 100% of windows — an unambiguous, genome-wide signature of population expansion (table 3, fig. 4, fig. 6). A moments fit to the folded SFS gives a growth history from an ancestral $N_e = 547,928$ to a present $N_e = 6,808,096$ over 769,519 generations (~ 12 -fold; generations $\approx$ years for this annual species), and this history reproduces both the observed π and a full-SFS Tajima’s $D = -1.92$ under forward simulation, ruling out a haploid/diploid scaling error. Stairway Plot 2 and GONE2 concur on strong recent expansion to N_e in the $1.5\text{--}1.7 \times 10^7$ range. Because sweepstakes reproduction would mimic expansion by inflating rare variants, we tested the coalescent class directly: the observed folded SFS is fit far better by a Kingman-plus-growth model than by any multiple-merger (Beta) coalescent (shape-deviation $L_1 = 0.036$ vs a best- α 0.108), and contemporary N_e estimates from the SFS and from linkage disequilibrium agree — both inconsistent with a strong multiple-merger process.

2.5 The recombination landscape

A deep-learning recombination-map method applied per sex gives an autosomal rate of $\sim 0.21\text{--}0.24$ cM/Mb (male 0.204–0.213, female 0.288–0.290, sex-averaged ~ 0.244 ; Supplementary Fig. S10). Because the estimator reads linkage disequilibrium, it recovers the sex-*averaged* rate, and heterochiasmy is not identifiable from it (a male/female autosomal split estimates the same quantity; the apparent gap was an artifact of unequal sample sizes). The Z chromosome shows a sex-biased rate, and a sex-averaged, boundary- corrected genome map underlies the sweep-scan calibration.

2.6 Diversity and gene-linked background selection

Nucleotide diversity is reduced in gene-dense regions by 20–30% (π correlates with local coding density at $r = -0.31$), a moderate background-selection effect that is gene-density-driven rather than recombination- driven (recombination is nearly uniform, section 2.5). A model-based B -map could not be fit under this regime (the classical approximation overflows where the deleterious mutation rate exceeds the selection coefficient), so we use local coding fraction as a background-selection covariate and control for it explicitly in the sweep scan (section 2.8). Because the callset is unphased, all diversity statistics are site-frequency-spectrum-based (π , Watterson’s θ , Tajima’s D); haplotype-homozygosity statistics are degenerate on these data and are not used.

2.7 Candidate adaptive genes (McDonald–Kreitman)

McDonald–Kreitman tests against the *I. coindetii* outgroup across 20,999 genes give a genome-wide proportion of adaptive substitutions $\alpha \approx 0.34$ –0.64. The single most strongly and significantly adaptive gene is the splicing factor *SF3B4* ($\alpha = 0.64$; a single-copy gene by our annotation). Per-gene power is limited — a genome-wide α of this size resolves cleanly to only a handful of genes — so we treat the MK results as complementary to, rather than overlapping with, the recent-sweep scan below, which detects a different (recent, often non-coding) timescale of selection. The second congener used here, *I. argentinus*, is a striking comparator in its own right: it is recently diverged from *Illex illecebrosus* (roughly an order of magnitude fewer fixed coding differences than *I. coindetii*; Supplementary Table S3) yet carries near-identical genome-wide diversity ($\pi = 0.0092$ by genotype-likelihood estimation on the same accessible sites, versus 0.0093 in *Illex illecebrosus* — within 1%) — two large- N_e , expansion-shaped congeners of matched standing variation, a comparison the single-species differentiation survey could not make.

2.8 A genome-wide scan for selective sweeps

We scanned for recent positive selection with a five-class site-frequency/diversity-based deep-learning classifier (neutral, hard, soft, linked-hard, linked-soft; 1.1 Mb feature windows of eleven 100 kb sub-windows), trained on forward and coalescent simulations under the fitted growth demography (fig. 5).

Two methodological findings shape the scan. First, a training/scanning feature mismatch — the model was initially trained on phased-haplotype simulation features but applied to unphased genotypes — inflated linkage-based features by an order of magnitude and had to be corrected by retraining on the same unphased feature extractor used to scan; the final full-SFS model reaches 81% class accuracy. Second, and more fundamentally, even the corrected model *over-calls* sweeps on the real genome: real neutral windows (accessible, ≥ 5 kb from any gene) carry 3–8-fold more *within-window spatial heterogeneity* in π and θ than the simulations, even though their marginal SFS matches ($D \approx -2$). We ruled out missing-data structure and sample incompleteness as the cause; the residual is real per-sub-window diversity variation (mutation-rate variation plus background selection) that a uniform- θ coalescent does not reproduce. We therefore use the classifier as a **stratified empirical-outlier detector** — ranking a sweep score ($p_{\text{hard}} + p_{\text{soft}}$) within strata of callable fraction and coding density so that a window is an outlier only relative to windows of similar accessibility and gene density — corroborated by a second, LD-footprint method. We also show that an inherited minor-allele-frequency filter was a mistake under strong expansion (it discards the rare-variant excess that carries both the expansion and the sweep signal) and move to the full spectrum on both the simulation and empirical sides.

Two tiers of sweep calls. We report sweeps at two levels of stringency (Supplementary Fig. S13).

Tier 1 is the diploSHIC hard/soft classification itself, spatially resolved by a recombination-aware Markov decoder — a hidden Markov model over the per-100 kb-window classes whose neutral/linked/centre emissions are the prior-corrected class posteriors and whose dwell lengths follow the ~ 400 kb sweep footprint scaled by the ReLERNN recombination map (Supplementary Fig. S13a). Because the genome reads as $\sim 91\%$ linked-selected under strong expansion, the decoder’s role is spatial coherence: it keeps hard/soft calls that lie in contiguous sweep centres and discards spatially isolated ones. This reduces the raw calls from 1,758 to 717 windows (658 localised calls; 459 hard, 199 soft; 35 on chromosomes 2 and 42), retaining the higher-outlier windows (median

empirical-outlier percentile 0.98 vs 0.95 for discarded calls; Supplementary Fig. S13b). **Tier 2** is the subset independently corroborated by an LD/SFS-footprint method (RAiSD or SweepFinder2): the 34 cross-method-concordant regions (the diploSHIC-HMM outlier plus at least one footprint method). The two tiers agree strongly — 89% of windows at concordant loci are decoded as sweep centres versus 2.9% genome-wide (odds ratio 282, $p < 10^{-46}$), and 30 of the 34 concordant regions contain a Tier-1 centre (Supplementary Fig. S13c) — so the conservative concordance set is a high-purity subset of the classifier calls, not a disjoint list. Because the decoder could not be simulation-calibrated at this genome’s recombination density (a dense-ancestry bottleneck; see Methods), its dwell parameters are set from the known sweep footprint and the recombination map and validated post-hoc by this concordance; we therefore treat Tier 1 as a ranked candidate set and Tier 2 as the high-confidence calls. The Z chromosome, which RAiSD and SweepFinder2 do not cover, is necessarily diploSHIC-only; it is scanned and localised separately below and contributes four further Tier-1 calls (genome-wide total 662), now shown alongside the autosomes in the genome-wide sweep figures (fig. 5; Supplementary Fig. S17).

The Tier-1 set is enriched for neural-developmental and RNA-regulatory function. Independently of the cross-method concordance, the 658 autosomal Tier-1 calls overlap 894 genes (476 with GO annotation; one-third of the calls fall in gene-poor, often transposon-rich windows and are excluded from the enrichment). A gene-ontology analysis (Fisher’s exact test, Benjamini–Hochberg FDR, against the annotated-gene background on the scanned chromosomes) returns 124 terms at FDR < 0.05 , dominated by two coherent themes: **neural development** (neurogenesis, neuron differentiation, cell-projection and axon morphogenesis; 1.5–1.8-fold) and **transcriptional/RNA-regulatory control** (regulation of RNA metabolic and biosynthetic process, transcription). The enrichment is carried entirely by the hard-sweep calls (99 enriched terms); the disjoint soft-sweep set has none at FDR < 0.05 and is underpowered. These themes recover and statistically reinforce the smaller Tier-2 concordant set — whose genes (the apelin receptor, *ZEB2*, *Rab3*, *cept1*) fall squarely within them — indicating that the recent-sweep target in *Illex illecebrosus* is a neural-developmental and gene-regulatory program, as expected for a behaviourally and neurally complex, fast-developing cephalopod. Screening the Tier-1 calls by background selection — the same per-region test used for the concordant set, retaining only calls whose local π falls below the background-selection expectation given flanking coding density — keeps **291 of the 658** (237 hard, 54 soft). The enrichment not only survives this filter but *strengthens*: 152 terms at FDR < 0.05 , with the neural-development and RNA-regulatory folds rising throughout (e.g. generation of neurons 1.6 $\rightarrow$ 1.9, transcription by RNA polymerase II 1.7 $\rightarrow$ 2.1), and a **photoreception** signal (response to light and radiation, ~ 3 -fold) emerging — a coherent sensory-and-neural adaptive program that is not an artefact of reduced diversity in gene-dense regions.

Candidate sweeps. The scan yields 216 candidate windows and **34 regions with cross-method concordance**; a per-region background-selection test (comparing local π to that expected from flanking coding density) isolates **11 high-confidence sweep candidates** in which diversity is reduced beyond the background-selection expectation. These carry a biologically coherent set of cephalopod-orthologous genes spanning neural and developmental function (*Rab3*, *ZEB2*), cardiovascular/vasculature signalling (*Aplnr*), genome-maintenance and stress response (*MACROD2*), and membrane/sphingolipid metabolism (*cept1*; and, at a locus the annotation had missed, a *ceramide phosphoethanolamine synthase*) — functions consistent with a fast, migratory, cold-water, neurally complex, short-lived animal (table 4). The single most distinctive autosomal candidate is a **cephalopod-lineage-specific gene under a hard sweep** on chromosome 16 (a tandem-exon

transcript with no functional annotation whose direct translation matches an uncharacterised protein family in the bobtail squid *Euprymna*, $E = 2 \times 10^{-53}$). Fine-scale localisation confirms all 11 candidates as recent sweeps by the sign of Tajima’s D at the diversity valley, and reassigns two apparent gene-desert signals to transposable-element regions (flagged as lower confidence) and one candidate’s target to a neighbouring gene. There is no overlap between the sweep candidates and the MK-adaptive genes, as expected for their different timescales.

The Z chromosome. Because males are Z-diploid and females Z-haploid, the Z was scanned separately in the 330 confident ZZ males, using a model retrained at the matched sample size (the classifier’s features are sample-size dependent) and a formal adequacy gate to control for the Z’s reduced effective size; the male-only screen was independently validated against the 289 haploid females, whose per-window diversity tracks the males’ at $r = 0.996$. Of the four Z Tier-1 calls (three hard, one soft; Supplementary Fig. S17), three credible, gene-localised sweeps survive fine-scale triage, all confirmed as conserved cephalopod genes by direct homology: **Pasilla/Nova-1** (a neuronal alternative-splicing regulator; 92% identity to *Octopus*), **Smaug/samd4a** (a germline RNA-binding translational repressor), and **Esrrg** (an estrogen-related receptor, with an adjacent likely-*cis*-regulatory lncRNA; 97.5% identity to *Octopus*) — a coherent RNA-regulation-and-hormone-signalling signature for a sex chromosome. Notably, the *Amnionless*/"Zfest" sex-determination locus falls just outside the swept valley and shows only purifying selection: it is a linked passenger, not the target.

The inversion is not a sweep outlier. Completing the scan across all 45 autosomes plus the Z, chromosome 2 yields *zero* concordant sweeps: the recombination-map method is broadly elevated across the inversion and the classifier over-calls there, but the two do not co-localise into focal sweep concordance — exactly the pattern expected for a balancing-selected supergene rather than a swept region, and correctly filtered by the concordance requirement.

3 Discussion

Illex illecebrosus emerges as a large- N_e , expansion-dominated system whose genome carries a single, dramatic axis of structure — a balancing-selected chromosome-2 supergene — against an otherwise unstructured, weakly background-selected background. Where an earlier survey (Baker et al., 2025) described this structure, our contribution is to explain it and to look past it: to show that the chr2 inversion is a selected, datable supergene with a specific frequency history, and to resolve the genome-wide sweep landscape — including on the sex chromosome — that a differentiation scan does not reveal. Three results deserve emphasis.

The inversion is the clearest case. Its rare, derived arrangement is more diverse than its frequency allows, which forces a history in which it was once common and is now declining; two mutation-rate-free arguments and a barrier-aware coalescent fit converge on an ~ 850 ky-old polymorphism, coincident with the onset of the population expansion, that has been actively removed over the last ~ 50 – 100 ky. That the age survived a complete reversal of the arrangement polarisation — because it is carried by the polarisation-symmetric divergence between arrangements — is a useful robustness check that few inversion-age estimates can claim. What the data do not resolve is the mechanism of balancing selection or the identity of its target: the block is gene-poor and differentiated as a unit, with no single locus standing out, consistent with selection on a co-adapted arrangement rather than a single gene.

The demographic result matters methodologically as much as biologically. A sweepstakes-reproduction species is exactly where one expects multiple-merger coalescents to masquerade as expansion; testing the coalescent class directly, rather than assuming Kingman, is what licenses every downstream age and selection estimate here.

Finally, the sweep scan is a cautionary and constructive example. Simulation-trained per-window classifiers are increasingly standard, but on a strongly expanding, unphased, missing-data-rich genome they over-call because real neutral sequence is spatially more heterogeneous than tractable neutral simulations. Rather than hide this behind a threshold, we quantified it and let it dictate the framing: a stratified empirical-outlier scan with cross-method concordance and an explicit background-selection filter. The resulting candidate set is small and biologically coherent — neural, RNA-regulatory, membrane-lipid, and stress-response genes, a lineage-specific novelty under a hard sweep, and a sex-chromosome cluster of RNA-regulation and hormone-signalling loci — and is offered as hypothesis-generating rather than as a list of confirmed targets.

4 Limitations

The callset is unphased, so no haplotype-homozygosity statistics are available; sweep detection relies on the SFS and diversity alone. The simulation-to-real neutral mismatch that drives classifier over-calling is real and unresolved at the level of absolute neutral calling — our empirical-outlier framing is a principled response, not a cure — and it also defeats a within-arrangement SFS approach to the inversion. Every inversion age scales inversely with the mutation rate, and $\mu = 3 \times 10^{-9}$ is the weakest external input; an independent calibration (e.g. a karyotyped congener) is the clearest route to a mutation-rate-free age. The corroborating LD-footprint method is broad and the abandoned frequency-spectrum method was unreliable on these data. Candidate sweep genes rest on homology to distant cephalopod models and on cross-method concordance; they are candidates, not confirmed adaptive targets.

5 Methods

Full parameter values, software versions, and command-level provenance are in Supplementary Table S1 and in the project repository.

Data. The ingroup is the 350-individual *Illex illecebrosus* subset of the baker_2025 whole-genome callset (of 633 sequenced), mapped to the *Illex_F24* primary assembly (45 numbered chromosomes plus the Z). Genotypes are unphased and the VCF is variants-only (no invariant sites), so diversity is computed from genotype likelihoods off the BAMs rather than from called genotypes. *I. coindetii* is the divergence/MK outgroup. An accessibility mask restricts all analyses to callable sites.

Diversity, SFS, and demography. Site-frequency spectra, Watterson’s θ , per-window π and Tajima’s D , and per-karyotype spectra were computed with ANGSD (-doSaf/realSFS). Demographic history was inferred with Stairway Plot 2, moments, and GONE2, and validated by forward simulation reproducing the observed π and D . Windowed π , d_{xy} , F_{ST} , and PCA used pg_gpu.

Karyotyping and the inversion. Karyotypes were called by local-PCA/lostruct and a Gaussian-mixture model. Arrangement polarisation used an AnchorWave *I. coindetii*↔*Illex illecebrosus* whole-genome alignment restricted to non-reference-biased sites. The inversion’s age and frequency trajectory were fit with msinv v0.3.5, an ARG-based coalescent simulator that enforces

cross-karyotype recombination barriers and outputs tree sequences, implementing the hulls-based ancestral-material tracking of Kelleher et al. (2016); it was validated against SLiM, msprime, and discoal. **All coalescent simulations underlying the inversion age and decline inference were generated with msinv**, and the complete chr2-inversion analysis — the fitting code, the simulation configuration, and every per-analysis output and figure — is housed as a dedicated *Illex illecebrosus* module within the msinv repository (`files/illex/`, `results/illex/`). Simulations used the fitted growth demography, $\mu = 3 \times 10^{-9}$, and $r = 1.977 \times 10^{-9}$; statistics were computed on the differentiated body and rescaled from a 37-kb simulated inversion with length-invariance verified over a 25-fold range. Breakpoints were pinned by combining arrangement differentiation with the AnchorWave alignment; missed genes were recovered by DIAMOND `blastx` against seven cephalopod proteomes, with transposable elements annotated by EarlGrey and transcripts by StringTie.

Recombination map. Per-sex recombination maps were built with ReLERNN (SIMULATE→TRAIN→PREDICT) dropping individuals with no calls on a chromosome and never forcing diploid genotype calls.

Selective-sweep scan. The classifier was diploSHIC (five classes, 1.1 Mb windows of eleven 100 kb sub-windows, twelve diploid statistics per sub-window), trained on SLiM 5.2 forward simulations (recapitated with msprime) and pure-msprime neutral simulations under the fitted growth demography, at the empirical sample size and with realistic structured missingness. The model was trained on the same unphased feature extractor (`fvecVcf`) used to scan, on the full spectrum (no MAF filter). It was used as a stratified empirical-outlier detector (score $p_{\text{hard}} + p_{\text{soft}}$ ranked within callable-fraction $\times$ coding-density strata), corroborated by RAISeD; SweepFinder2 and a *B*-value calculator were evaluated but not used for reported results. The Z chromosome was scanned in males only with a model retrained at $n = 330$ and passed a comparative feature-adequacy gate, then validated against the haploid females. Candidate genes were annotated with EnTAP and screened by DIAMOND against cephalopod proteomes; a per-region background-selection test compared local π to the flanking-coding-density expectation. The per-window classifications were spatially resolved into Tier-1 calls with a recombination-aware hidden Markov model (neutral / linked / sweep-centre states; emissions are the prior-corrected collapsed class posteriors; dwell lengths set from the ~ 400 kb sweep footprint scaled by the ReLERNN sex-averaged map, with a constant-rate fallback on chromosomes 2 and 42). A simulation calibration of the decoder was attempted but proved infeasible on this genome: at N_e -scale recombination the recapitated ancestral recombination graphs carry $\sim 3.4 \times 10^5$ marginal trees per 2.5 Mb, making the tree-sequence genotype ascertainment for the training scan prohibitively slow (> 2.7 h per segment). The decoder’s dwell parameters are therefore design-informed from the sweep footprint and the recombination map rather than fit to simulations, and are validated post-hoc by the 89%/2.9% Tier-1–Tier-2 concordance (Supplementary Fig. S13).

McDonald–Kreitman. MK tests used mkado against the *I. coindetii* outgroup over 20,999 genes.

Data and code availability

The analysis code, per-step results, and the working project document (`ILLEX_PROJECT_MANUSCRIPT_AND_METHODS`) are in the project repository. The coalescent simulations for the inversion analysis (section 2.3) were generated with `msinv` v0.3.5, a general ARG-based inversion-coalescent simulator ([https:](#)

Table 1: **Sample and whole-genome summary.** Diversity statistics are genome-wide from genotype likelihoods; karyotype counts and the inverted-arrangement frequency are for chromosome 2.

Quantity	Value
Species / callset	<i>Illex illecebrosus</i> ; baker_2025 whole-genome
Individuals (karyotyped)	633
Sex (chrZ coverage)	342 ZZ males / 291 ZO females
Sampling	11 NAFO divisions, $\sim 37\text{--}50^\circ\text{N}$
Geographic structure	$F_{ST} \approx 0$ (none outside chr2)
Genome-wide π	0.0093
Genome-wide Tajima’s D	-2.07 (-2.09 in 50 kb windows)
chr2 karyotypes (AA/AB/BB)	254 / 284 / 95
Inverted (BB) frequency	0.374 (0.29–0.46 across divisions, non-clinal)
Karyotype $F_{ST}(\text{AA}, \text{BB})$	0.365

Table 2: **The chromosome-2 inversion** (corrected polarisation: I = BB inverted/derived, S = AA ancestral/standard).

Quantity	Value
Location (differentiated body)	chr2:60.54–79.50 Mb (~ 18.96 Mb)
Karyotypes (AA/AB/BB)	254 / 284 / 95 (95.3% confident)
Inverted (BB) frequency p	0.374
Historical peak frequency	~ 0.74
Karyotype $F_{ST}(\text{AA}, \text{BB})$	0.365 (geographic $F_{ST} \approx 0$)
π_I/π_S (I=BB)	1.356 ± 0.048
d_{xy}/π_I	1.385 ± 0.021
Coalescent age t_{inv}	~ 850 ky ($\propto 1/\mu$, $\mu = 3 \times 10^{-9}$)
Decline	$0.74 \rightarrow 0.374$ over $\sim 50\text{--}100$ ky, ending $\gtrsim 50\text{--}100$ ka
Selection on decline	$N_e s \sim 10^2\text{--}10^3$ ($s \approx 1.5 \times 10^{-4}$)

[//github.com/stsmall/msinv](https://github.com/stsmall/msinv), MIT); it is not specific to *Illex illecebrosus*, but this study’s complete *Illex illecebrosus*-specific inversion analysis — fitting code, simulation configuration, tabulated outputs, and the inversion figures — is contained within that repository under `files/illex/` and `results/illex/`. A section-by-section index of every figure and table source file is provided as `FIGURES_AND_TABLES_INDEX.md`.

Acknowledgements

[TODO: funding, sequencing, sample collection, computing.]

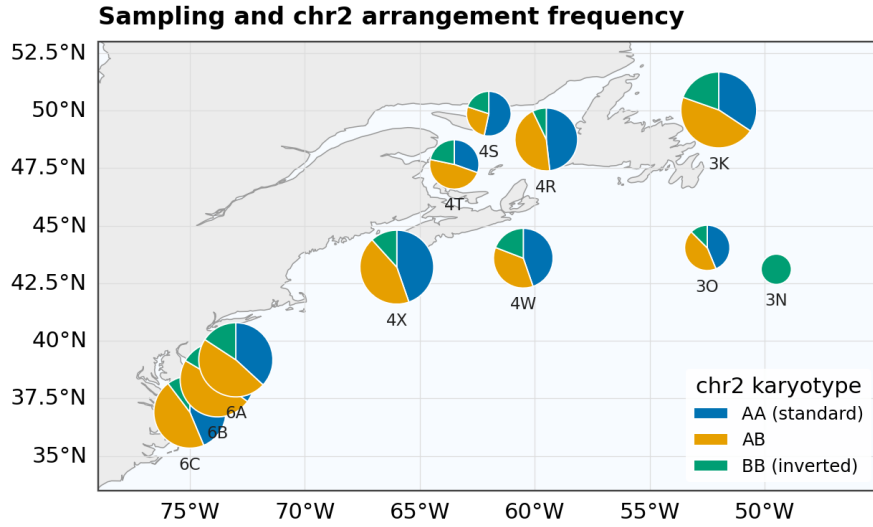


Figure 1: **Sampling and chromosome-2 arrangement frequency.** *Illex illecebrosus* sampled across eleven NAFO divisions ($\sim 37\text{--}50^\circ\text{N}$); pies show the AA/AB/BB karyotype composition at each division (sized by N). The inverted (BB) frequency is non-clinal (0.29–0.46 across latitude), consistent with geographic $F_{ST} \approx 0$.

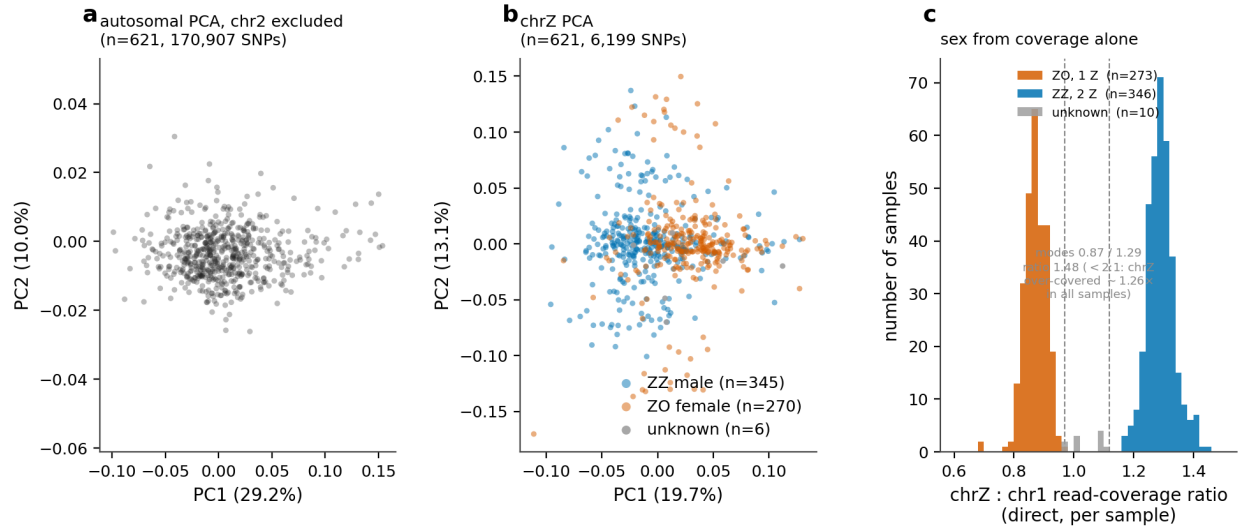


Figure 2: **No population structure outside chr2, and the sex assignment** (recapitulating Baker et al. 2025). (a) Genome-wide autosomal PCA with chr2 excluded — a single undifferentiated cloud. (b) chrZ PCA (mild male/female separation). (c) chrZ:autosome coverage ratio (chr42-corrected for a shared mapping bias) by sex: ZZ males at ~ 1.0 (two Z copies); ZO females clearly lower, near the single-copy expectation (there is no W) — the residual offset above 0.5 is most likely a low-coverage calibration artifact rather than biology. 342 males / 291 females.

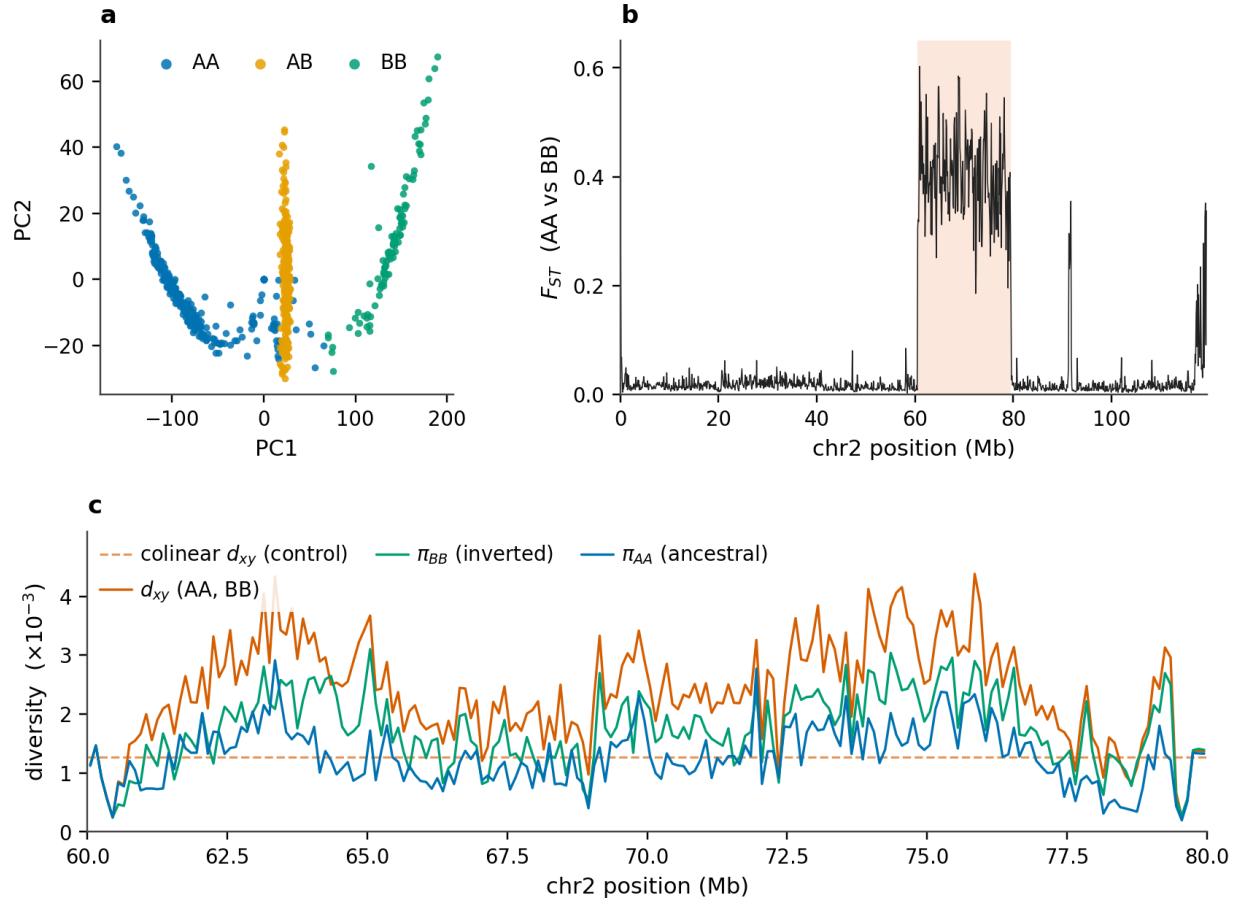


Figure 3: **The chromosome-2 inversion.** (a) chr2 PCA resolves three karyotype clusters (AA/AB/BB). (b) F_{ST} between arrangements along chr2 — the inversion (60.54–79.50 Mb, shaded) is the only differentiated region (median F_{ST} = 0.37 vs 0.01 collinear). (c) Per-arrangement diversity across the inversion: the derived, inverted BB arrangement is *more* diverse than the ancestral AA (π_I/π_S = 1.36), and d_{xy} exceeds both.

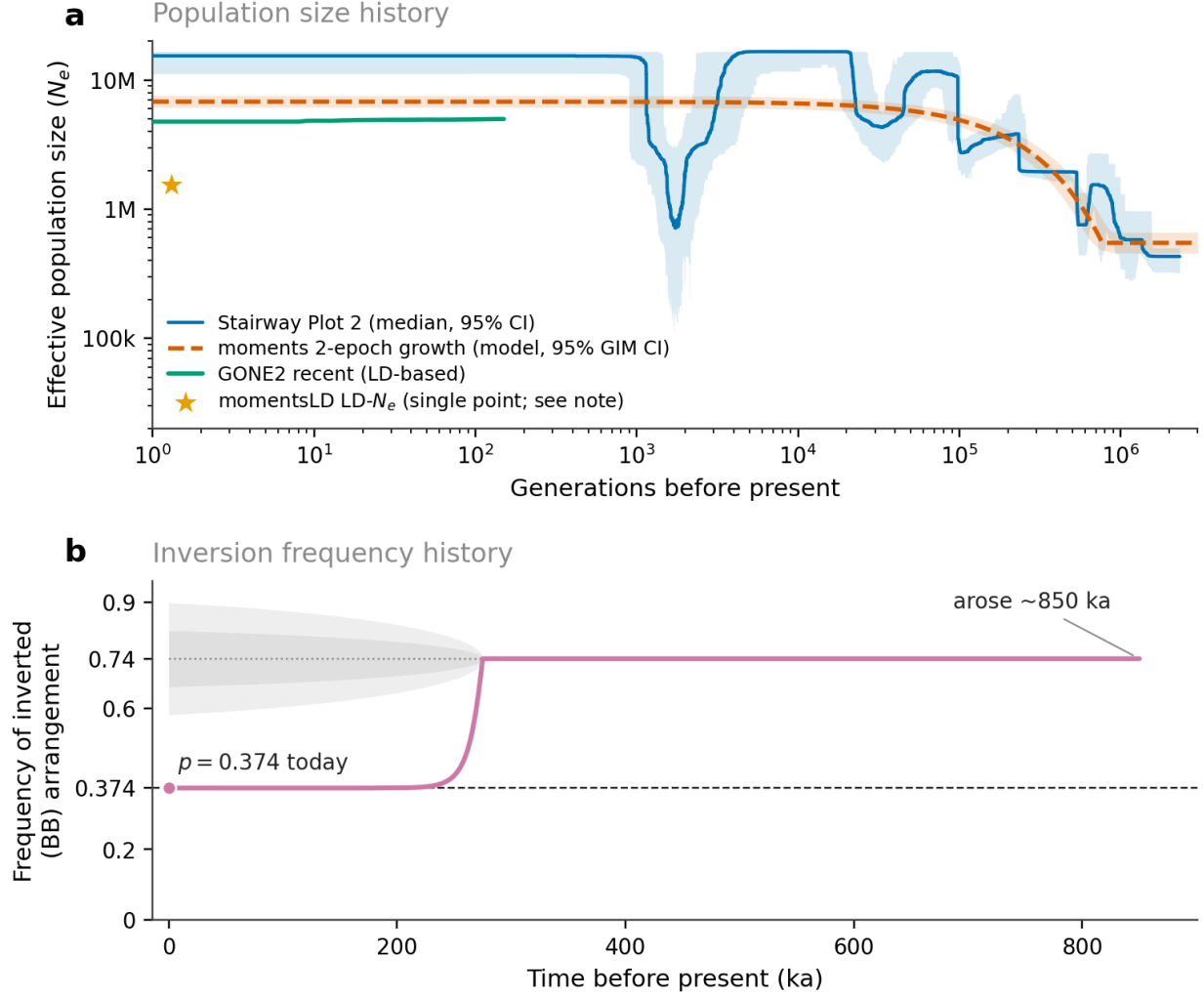


Figure 4: **Demographic history and the inversion's frequency history.** (a) Effective population size over time: Stairway Plot 2 (median $\pm 95\%$ CI), the moments 2-epoch growth model with its Godambe (GIM) 95% parameter-CI envelope (expansion $12.4 \times \pm 9\%$, $T \approx 770$ ky $\pm 16\%$), GONE2 for the recent past, and a single momentsLD LD- N_e point (rate-corrected; only the contemporary point is shown because its time-varying fit is non-identifiable). (b-i) Fitted inversion trajectory: the derived (BB) arrangement arose ~850 ka, was held near $p \approx 0.74$, and declined to 0.374, with the neutral-drift envelope. (b-ii, b-iii) $\Delta\chi^2$ profiles locating the end (t_{decline}) and duration (t_{fall}) of the decline; the fall was recent and fast, excluding neutral drift-out.

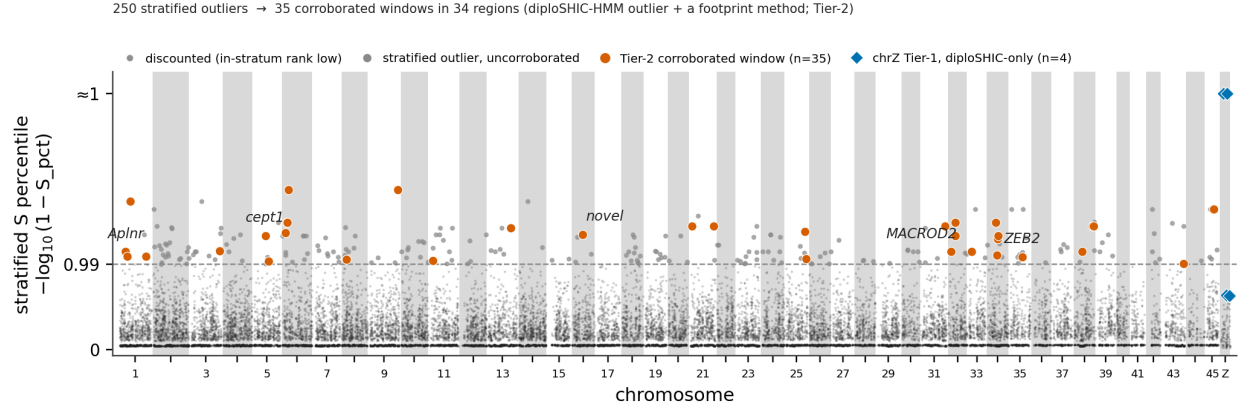


Figure 5: **The genome-wide selective-sweep scan.** Sweep signal across the genome (chromosomes concatenated). The classifier reads the genome as pervasively *linked*-selected ($\sim 91\%$ linked-hard/soft; neutral $\sim 1\%$; true hard/soft only a few percent), so raw diploSHIC calls are filtered by cross-method concordance (a diploSHIC-HMM outlier plus RAiSD or SweepFinder2) and a background-selection test; the corroborated, background-selection-robust candidates (35 windows across 34 regions; highlighted) stand out above the discounted background, with key genes labelled. Chromosome-Z Tier-1 calls (diamonds) are male-based and diploSHIC-only — RAiSD/SweepFinder2 do not cover the Z — so they cannot enter the concordance filter. The full scan pipeline is shown in Supplementary Fig. S6.

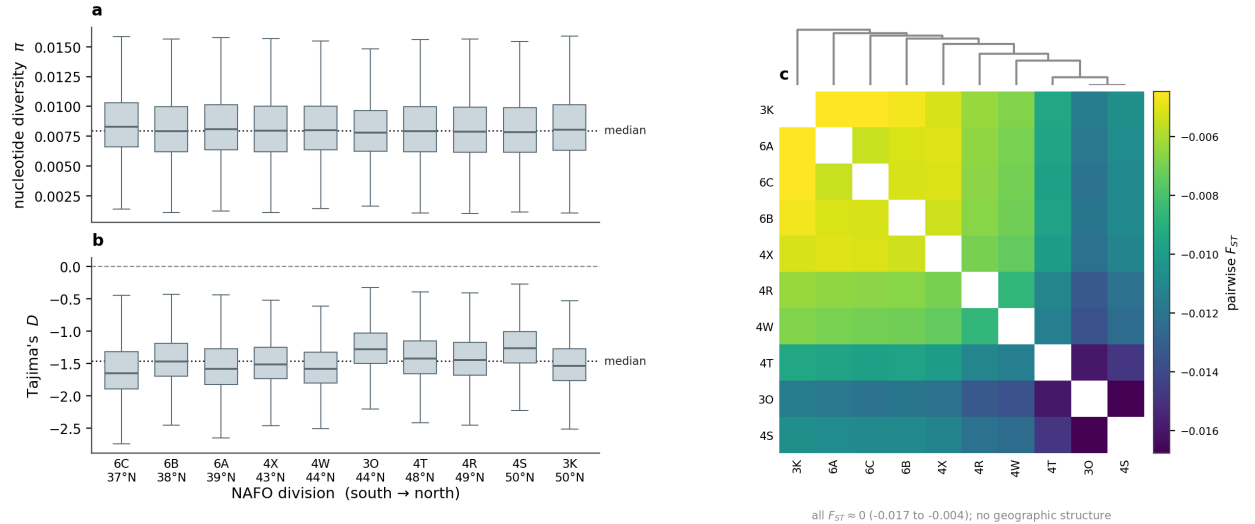


Figure 6: **Diversity is uniform and there is no geographic structure.** (a) Nucleotide diversity π and (b) Tajima's D per NAFO division, ordered south→north (coloured by latitude) — both are flat across latitude (no cline), and D is negative in every population, the signature of strong expansion. (c) Pairwise F_{ST} among divisions as a clustered heatmap with dendrogram: every value is ≈ 0 (-0.017 to -0.004) and the clustering follows sample size, not geography — no population structure. (π/D estimated by ANGSD genotype likelihoods, large divisions subsampled to $n = 25$; F_{ST} genome-wide by pg_gpu. Per-chromosome distributions: Supplementary Fig. S7.)

Table 3: **Demographic model** (moments, folded SFS; validated by forward simulation).

Parameter	Value
Ancestral N_e (N_REF)	547,928
Present N_e (N0)	6,808,096
Growth duration (T_GROW)	769,519 generations
Fold expansion	$\sim 12\times$
Genome-wide π	0.0093
Genome-wide Tajima’s D	≈ -2 (nearly all windows)
Mutation rate μ	3×10^{-9} / bp / generation
Recombination (autosomal)	$\sim 0.21\text{--}0.24$ cM/Mb (sex-averaged)

Table 4: **High-confidence selective-sweep candidates**, grouped by biological function and ordered by chromosome, including the Z-chromosome set. Autosomal rows are the 11 background-selection-robust candidates (of 34 concordant regions); Z rows (marked) are the fine-scale-confirmed sweeps from the males-only scan. Genes are cephalopod orthologs. Class: hard/soft sweep. Full machine-readable tables: `outlier_scan/` (Supplementary Table S3).

Location	Gene / feature	Class
<i>Neural, RNA-regulation & development</i>		
chr35:39.0 Mb	<i>ZEB2</i> (neural/EMT transcription factor)	hard
Z:20.8 Mb	<i>Pasilla/Nova-1</i> (splicing regulator)	hard (Z)
Z:23.6 Mb	<i>Smaug/samd4a</i> (translational repressor)	hard (Z)
<i>Cardiovascular</i>		
chr1:20.3 Mb	<i>Aplnr</i> (apelin receptor)	hard
<i>Genome-maintenance & stress</i>		
chr32:9.3 Mb	<i>MACROD2</i> (DNA-damage response)	soft
<i>Membrane & lipid</i>		
chr5:38.0 Mb	<i>cept1</i> (phospholipid synthesis)	hard
chr1:85.5 Mb	ceramide phosphoethanolamine synthase [†]	hard
<i>Metabolism</i>		
chr25:65.0 Mb	<i>GALT</i> [†]	soft
<i>ECM & adhesion</i>		
chr33:9.0 Mb	<i>Tinagl1</i>	hard
<i>Signalling</i>		
chr32:24.1 Mb	GPCR cluster	hard
Z:30.7 Mb	<i>Esrrg</i> (estrogen-related receptor)	soft (Z)
<i>Cephalopod-specific novelty</i>		
chr16:35.3 Mb	uncharacterised (<i>Euprymna</i> family)	hard
<i>Lower confidence (TE-dense)</i>		
chr1:35.9 Mb	transposable element [‡]	hard
chr11:10.8 Mb	transposable element [‡]	hard

[†] target reassigned by fine-scale localisation (annotation-recovered gene). [‡] TE-dense region; deep- π /negative- D may be a TE-insertion sweep or a repeat artifact.

References

- Baker et al. [CONFIRM — the Baker et al. *Illex illecebrosus* population-genomics paper: PCA/FST/admixture, chr2 inversion structure; the callset source for this study]. *[journal]*, 2025. CONFIRM full citation.
- Jerome Kelleher, Alison M. Etheridge, and Gilean McVean. Efficient coalescent simulation and genealogical analysis for large sample sizes. *PLoS Computational Biology*, 12(5):e1004842, 2016.